CLASS 10 · JEE ADVANCED · TOP-10 QUESTION IMPACT REPORT

Pragya G10 (EXI) Achievers

Part Test-1 · Paper 2 · which questions cost the top 10 the most

Exam date 09 Oct 2026
Campus HYD/Nallakunta DD Colony
Students analysed top 10 by program rank

In total, the top 10 lost **≈ 463 marks** – **≈ 181** from Maths, **≈ 186** from Physics and **≈ 96** from Chemistry.



Marks lost = marks not earned on wrong or skipped questions plus negative marks. Approximate (≈) because multiple-correct questions give partial marks.

11 of 50
high-impact questions – lost by 5+ of the top 10

40%
of all lost marks came from Physics

Q7 8/10
most-lost question – plus 3 more questions at 8/10

129 / 180
top-10 average score (range 118–164)

1 Where the damage happened

Marks the top 10 gave away in each subject, split into marks lost on **wrong answers** (the question's marks plus the negative mark) and marks left on **skipped** questions. Advanced figures are approximate (≈) – see note at the end.

SUBJECT	SPLIT ■ wrong ■ skipped	MARKS LOST	SHARE	ACCURACY*	HIGH-IMPACT
Maths	≈ 142 on 31 wrong answers (incl. 29 negative) · 39 on 11 skipped	**≈ 181** ≈ 18 per student	39%	81%	**3** of 17 Qs
Physics	≈ 150 on 42 wrong answers (incl. 11 negative) · 36 on 12 skipped	**≈ 186** ≈ 19 per student	40%	72%	**6** of 16 Qs
Chemistry	≈ 81 on 22 wrong answers (incl. 4 negative) · 15 on 4 skipped	**≈ 96** ≈ 10 per student	21%	87%	**2** of 17 Qs

* Accuracy = right answers ÷ questions attempted.

BY QUESTION TYPE	MATHS	PHYSICS	CHEMISTRY
Single correct marking +3 / −1	Q1–4 **12%** wrong · **10%** skipped	Q18–21 **22%** wrong · **12%** skipped	Q35–38 **5%** wrong · **2%** skipped
Multiple correct marking +4 / −2	Q5–7 **40%** wrong · **7%** skipped	Q22–24 **3%** wrong · **0%** skipped	Q39–41 **3%** wrong · **0%** skipped
Integer marking +4 / 0	Q8–13 **13%** wrong · **7%** skipped	Q25–30 **24%** wrong · **0%** skipped	Q42–47 **17%** wrong · **5%** skipped
Paragraph marking +3 / 0	Q14–17 **15%** wrong · **2%** skipped	Q31–34 **50%** wrong · **18%** skipped	Q48–51 **22%** wrong · **0%** skipped

Key takeaways

1. **Physics** did the most damage: **40%** of all marks lost by the top 10 (≈ 186 of ≈ 463). 6 of its 16 questions were lost by 5 or more students.
2. Biggest chapter-level loss: **Physics · Kinematics** – ≈ 110 marks across 8 questions (3 high-impact).
3. Skipping is concentrated in **Physics · Paragraph**: 7 of 40 answers (18%) were left blank.
4. Negative marking alone cost the top 10 **44 marks** (about 4 per student).
5. Strong ground: all 10 answered **15 questions** correctly (Q3, Q9, Q10, Q22, Q23, Q26, Q30, Q35, Q37, Q38, Q39, Q41, Q43, Q45, Q48).

Answered wrong by 5+ students

Students attempted these and still got them wrong – points to a misconception or a trap worth re-teaching.

Q7 8 wrong · **Q28** 5 wrong · **Q31** 5 wrong · **Q32** 5 wrong · **Q33** 5 wrong · **Q34** 5 wrong · **Q46** 5 wrong · **Q51** 5 wrong

Left blank by 5+ students

Most students did not attempt these – points to an unfamiliar topic, low confidence, or running out of time.

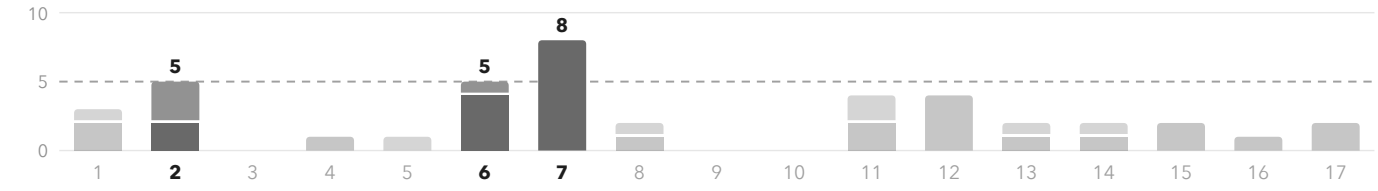
None

2 Question-by-question impact

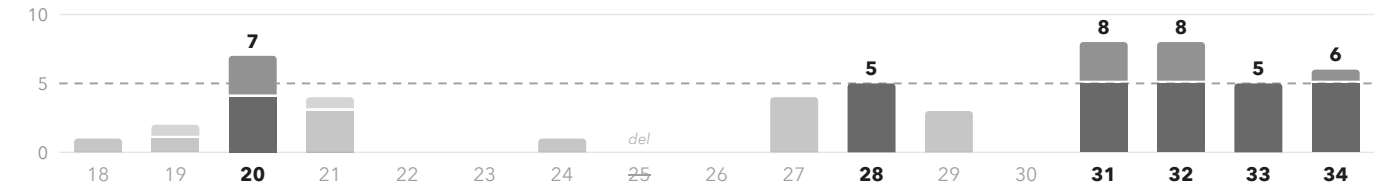
How many of the top 10 lost each question. Questions at or above the dashed line (lost by 5+) are **high-impact** and shown in full colour with their count.

■ Wrong answer ■ Skipped ---- High-impact line (5 of 10) ■ Below the line (faded) "del" = deleted / bonus question

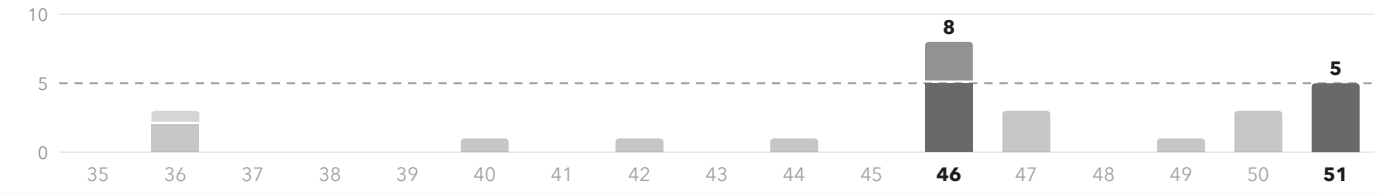
Maths — **3** of 17 questions high-impact · ≈ **181** marks lost by the top 10



Physics — **6** of 16 questions high-impact · ≈ **186** marks lost by the top 10



Chemistry — **2** of 17 questions high-impact · ≈ **96** marks lost by the top 10



3 Chapter hot-spots

Marks lost by the top 10, chapter by chapter. Subjects are ordered from most marks lost to least; within each subject, re-teach from the top.

ⓘ **Marks lost** adds up what all 10 students gave away on a chapter's questions: the question's marks plus its negative mark for each wrong answer (single correct 3 + 1 = 4, multiple correct 4 + 2 = 6, integer 4, paragraph 3), and the question's marks for each skipped one. Example: **Physics · Kinematics** has 8 questions, and the top 10 lost ≈ **110** marks on them — about **11 per student**. **Students per Q** = on average, how many of the 10 lost each question in that chapter (here 4.0 of 10).

CHAPTER	QS	HIGH-IMPACT QUESTIONS	MARKS LOST BY TOP 10	PER STUDENT	STUDENTS PER Q
1. Physics ≈ **186** marks lost · **40%** of all lost marks · **6** high-impact questions					
Kinematics	8	**Q20**, **Q31**, **Q32**	**≈ 110**	≈ 11 marks	4.0 of 10
Electrostatics	6	**Q28**, **Q33**, **Q34**	**≈ 72**	≈ 7 marks	3.5 of 10
Current Electricity	2	None reached 5 · most missed Q18 (1/10)	**≈ 4**	≈ 0 marks	0.5 of 10
2. Maths ≈ **181** marks lost · **39%** of all lost marks · **3** high-impact questions					
Differential Equations	5	**Q7**	**≈ 90**	≈ 9 marks	3.8 of 10
Definite Integration	5	**Q2**	**≈ 37**	≈ 4 marks	2.2 of 10
Area Under Curves	3	**Q6**	**≈ 31**	≈ 3 marks	2.0 of 10
Indefinite Integration	4	None reached 5 · most missed Q1 (3/10)	**≈ 23**	≈ 2 marks	1.5 of 10
3. Chemistry ≈ **96** marks lost · **21%** of all lost marks · **2** high-impact questions					
Carboxylic Acids	4	**Q51**	**≈ 40**	≈ 4 marks	3.0 of 10
Alcohols, Phenols & Ethers	7	**Q46**	**≈ 38**	≈ 4 marks	1.3 of 10
Aldehydes & Ketones	6	None reached 5 · most missed Q36 (3/10)	**≈ 18**	≈ 2 marks	0.8 of 10

4 High-impact questions at a glance

Every question lost by 5 or more of the top 10, one table per subject, worst first. The cropped question for each is in section 7.

Marks lost = marks the top 10 together gave away on that question: the question's marks plus its negative mark for each wrong answer (single correct 3 + 1 = 4, multiple correct 4 + 2 = 6, integer 4, paragraph 3), and the question's marks for each skipped one. Example: **Q7** – 8 wrong × 6 = 48 marks. **Lost by** = how many of the 10 got it wrong or skipped it.

Critical lost by 9–10 **High** lost by 7–8 **Moderate** lost by 5–6

Maths

3 of 17 questions high-impact · ≈ **93** marks lost on them by the top 10

Q	CHAPTER – WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q7	**Differential Equations** – Curve from tangent-intercept ratio BP : AP = 3 : 1	Multiple · +4 / −2	**8/10**	8	0	**≈ 48**
Q6	**Area Under Curves** – Line x = α halving the area between x^3 and x	Multiple · +4 / −2	**5/10**	4	1	**≈ 28**
Q2	**Definite Integration** – $f(a) = \int \ln(1+x^a)/(1+x)\,dx$ – find f(4) + f(1/4)	Single · +3 / −1	**5/10**	2	3	**≈ 17**

Physics

6 of 16 questions high-impact · ≈ **126** marks lost on them by the top 10

Q	CHAPTER – WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q31	**Kinematics** – Hexagon projectile – maximum height	Paragraph · +3 / 0	**8/10**	5	3	**≈ 24**
Q32	**Kinematics** – Hexagon projectile – speed at the top	Paragraph · +3 / 0	**8/10**	5	3	**≈ 24**
Q20	**Kinematics** – Shell fired from a plateau – minimum landing distance	Single · +3 / −1	**7/10**	4	3	**≈ 25**
Q34	**Electrostatics** – Flux through a cube from a wire along one edge	Paragraph · +3 / 0	**6/10**	5	1	**≈ 18**
Q28	**Electrostatics** – Flux through a cube with charges at its vertices	Integer · +4 / 0	**5/10**	5	0	**≈ 20**
Q33	**Electrostatics** – Flux through a ring – solid angle	Paragraph · +3 / 0	**5/10**	5	0	**≈ 15**

Chemistry

2 of 17 questions high-impact · ≈ **47** marks lost on them by the top 10

Q	CHAPTER – WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q46	**Alcohols, Phenols & Ethers** – Identifying polar aprotic solvents	Integer · +4 / 0	**8/10**	5	3	**≈ 32**
Q51	**Carboxylic Acids** – Moles of $LiAlH_4$ to reduce RCOOH to RCH_2OH	Paragraph · +3 / 0	**5/10**	5	0	**≈ 15**

5 Who lost which high-impact question

Each row is a student (campus order); each column a high-impact question. **Prog. rank** = rank in the program, **AIR** = All-India rank, **Total** = marks out of 180, **Avg** = average of the 6 weekly Advanced tests (WTA), out of 180. ✗ = wrong, – = skipped, ✓ = correct.

■ ✗ Wrong ■ – Skipped ■ ✓ Correct

	Rank & score				Maths			Physics						Chem		
STUDENT	PROG. RANK	AIR	TOTAL	AVG /180	2	6	7	20	28	31	32	33	34	46	51	Lost
Thulla Anuradhan Reddy	1	**1**	164	137.2*	✓	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	**2**
Dharshan Raj S	2	**17**	136	125.2*	✓	✓	✓	✗	✓	–	–	✓	✓	✗	✗	**5**
Thonta Harshavardhan	4	**26**	130	96.0*	✗	✓	✗	✗	✓	✗	✗	✓	✗	✗	✗	**8**
Poli Karthikeya Reddy	5	**30**	129	113.4	–	✓	✗	✗	✗	–	–	✗	–	✓	✗	**9**
K Ranveer Reddy	6	**31**	128	106.8	–	✓	✗	–	✗	✗	✗	✓	✗	✗	✓	**8**
Shiva S. J. Kokkonda	7	**36**	125	108.6*	✓	✗	✗	✓	✗	–	–	✗	✓	–	✗	**8**
Mano Nischal Reddy D	8	**42**	123	90.2*	–	✗	✗	–	✓	✗	✗	✓	✗	✗	✓	**8**
Vihaan Bussari	12	**50**	118	106.7	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗	✓	**8**
Gajavali Anjaneyulu	13	**52**	118	92.6*	✓	✓	✗	–	✗	✗	✗	✗	✗	–	✗	**9**
Yapala Dhanush Reddy	14	**55**	118	104.2	✓	–	✓	✗	✓	✓	✓	✗	✗	–	✓	**5**
Students who lost it					5	5	8	7	5	8	8	5	6	8	5	

6 Subject scores of the top 10

Estimated subject scores (≈) from right / wrong / skipped answers; partial credit on multiple-correct questions is not in the source data, so these can run a few marks above the official total. The weakest subject (lowest % of its maximum) is highlighted. **Subject maximum** leaves out the deleted / ungraded questions: Physics 56 = 60 minus Q25 (4).

STUDENT	PROG. RANK	AIR	TOTAL	AVG /180	**Maths** score · right / wrong / skipped	**Physics** score · right / wrong / skipped	**Chemistry** score · right / wrong / skipped	WEAKEST
Thulla Anuradhan Reddy	1	**1**	**164**	137.2*	≈ **48**/60 15 / 2 / 0	≈ **56**/56 16 / 0 / 0	≈ **60**/60 17 / 0 / 0	Maths
Dharshan Raj S	2	**17**	**136**	125.2*	≈ **56**/60 16 / 1 / 0	≈ **42**/56 12 / 2 / 2	≈ **40**/60 12 / 5 / 0	Chemistry
Thonta Harshavardhan	4	**26**	**130**	96.0*	≈ **46**/60 14 / 3 / 0	≈ **39**/56 11 / 5 / 0	≈ **45**/60 13 / 4 / 0	Physics
Poli Karthikeya Reddy	5	**30**	**129**	113.4	≈ **44**/60 13 / 3 / 1	≈ **32**/56 9 / 4 / 3	≈ **53**/60 15 / 2 / 0	Physics
K Ranveer Reddy	6	**31**	**128**	106.8	≈ **36**/60 11 / 4 / 2	≈ **36**/56 10 / 5 / 1	≈ **56**/60 16 / 1 / 0	Maths
Shiva S. J. Kokkonda	7	**36**	**125**	108.6*	≈ **45**/60 14 / 3 / 0	≈ **37**/56 10 / 2 / 4	≈ **46**/60 13 / 2 / 2	Physics
Mano Nischal Reddy D	8	**42**	**123**	90.2*	≈ **34**/60 11 / 5 / 1	≈ **36**/56 10 / 5 / 1	≈ **56**/60 16 / 1 / 0	Maths
Vihaan Bussari	12	**50**	**118**	106.7	≈ **44**/60 14 / 3 / 0	≈ **25**/56 8 / 8 / 0	≈ **49**/60 14 / 3 / 0	Physics
Gajavali Anjaneyulu	13	**52**	**118**	92.6*	≈ **43**/60 13 / 4 / 0	≈ **25**/56 7 / 8 / 1	≈ **50**/60 14 / 2 / 1	Physics
Yapala Dhanush Reddy	14	**55**	**118**	104.2	≈ **23**/60 7 / 3 / 7	≈ **46**/56 13 / 3 / 0	≈ **49**/60 14 / 2 / 1	Maths

Average. "Avg" = the student's average over the 6 weekly Advanced tests (WTA) before this Part Test, out of 180 (those tests are marked out of 240; averages are scaled to 180 to match this paper — × 180/240, percentages unchanged), from the campus Mains & Advance averages sheet. * Sat fewer weekly tests (tests sat / conducted): Thulla Anuradhan Reddy (5/6), Dharshan Raj S (5/6), Thonta Harshavardhan (5/6), Shiva Sai Jyothiraditya Kokkonda (5/6), Mano Nischal Reddy D (5/6), Gajavali Anjaneyulu (5/6).
Ranks. Program rank = rank within Pragya G10 (EXI) Achievers across all campuses (from the final marks sheet; the top 10 are picked by this rank). **AIR** = All-India rank across the whole Class 10 programme (all batches, all campuses), from the institute's All India Marks Analysis sheet.
How to read this report. "Lost" means the student did not get the question right — either answered it wrong or left it blank. A question is **high-impact** when 5 or more of the top 10 lost it. **"Marks lost"** = marks the top 10 together gave away: the question's marks plus its negative mark for each wrong answer (single correct 3 + 1 = 4, multiple correct 4 + 2 = 6, integer 4, paragraph 3), and the question's marks for each skipped one. "Per student" divides that by 10. **Deleted question:** Q25 has no right / wrong / skipped entry for any student in the final marks sheet (deleted / bonus question), so it is left out of all counts and shown as "del" in the charts. Chapter tags were assigned by reading each question; please correct any that your department classifies differently.
Note on Advanced marks. The paper uses partial marking for multiple-correct questions, but the source data only says right / wrong / skipped. Marks-lost figures use the paper's marking scheme (shown on every question) and are approximate; the counts of students who lost each question are exact.

7 Maths – high-impact questions

3 questions, worst first
cropped from the question paper

Q7 Differential Equations – Curve from tangent-intercept ratio BP : AP = 3 : 1

Maths · Multiple correct · marking +4 / −2 · marks lost by the top 10: 8 wrong × 6 = 48

Lost by 8/10
8 wrong · 0 skipped

7. A curve $y = f(x)$ passes through the point (1,1) and tangent at P(x,y) cuts the x-axis and y-axis at A and B, respectively such that BP:AP=3:1, then

A) Equation of the curve is $xy^3 = 1$

B) Equation of normal at the point (1,1) is $x - 3y + 2 = 0$

C) Curve passes through the point $\left(\frac{1}{8}, 2\right)$

D) Equation of the curve is $x\frac{dy}{dx} + 3y = 0$

Wrong: Thulla Anuradhan Reddy, Thonta Harshavardhan, Poli Karthikeya Reddy, K Ranveer Reddy, Shiva Sai Jyothiraditya Kokkonda, Mano Nischal Reddy D, Vihaan Bussari, Gajavali Anjaneyulu

Skipped: –

Q6 Area Under Curves – Line x = α halving the area between x³ and x

Maths · Multiple correct · marking +4 / −2 · marks lost by the top 10: 4 wrong × 6 + 1 skipped × 4 = 28

Lost by 5/10
4 wrong · 1 skipped

6. If the line x=α divides the area of the region $R = \{(x, y) \in R^2 : x^3 \le y \le x, 0 \le x \le 1\}$ into two equal parts, then

A) $\frac{1}{2} \le \alpha \le 1$ B) $\alpha^4 + 4\alpha^2 - 1 = 0$ C) $0 \le \alpha \le \frac{1}{2}$ D) $2\alpha^4 - 4\alpha^2 + 1 = 0$

Wrong: Thulla Anuradhan Reddy, Shiva Sai Jyothiraditya Kokkonda, Mano Nischal Reddy D, Vihaan Bussari

Skipped: Yapala Dhanush Reddy

Q2 Definite Integration – f(a) = ∫ ln(1+x^a)/(1+x) dx – find f(4) + f(1/4)

Maths · Single correct · marking +3 / −1 · marks lost by the top 10: 2 wrong × 4 + 3 skipped × 3 = 17

Lost by 5/10
2 wrong · 3 skipped

2. Let $f(a) = \int_0^1 \frac{\ln(1+x^a)}{1+x} dx$ then $f(4) + f\left(\frac{1}{4}\right)$ is equal to

A) $\ln^2 1$ B) $\ln^2 2$ C) $\ln^2 3$ D) $\ln^2 4$

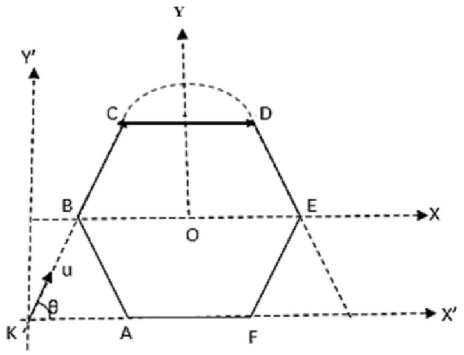
Wrong: Thonta Harshavardhan, Vihaan Bussari

Skipped: Poli Karthikeya Reddy, K Ranveer Reddy, Mano Nischal Reddy D

Q31 **Kinematics – Hexagon projectile – maximum height**
Physics · Paragraph · marking +3 / 0 · marks lost by the top 10: 5 wrong × 3 + 3 skipped × 3 = 24

Lost by 8/10
5 wrong 3 skipped

PARAGRAPH-I



A regular hexagon of side a=$10\sqrt{3}$ m is shown in figure. From any point a particle is projected such that it touches B,C,D and E. The velocity of particle is v m/s. O is the centre of the hexagon. x-axis is parallel to horizontal and y-axis is perpendicular to AE and CD (g=10 m/s^2) $\left(\sqrt{3}=1.73\right)$.

31. The maximum height from the horizontal surface X' attained by the projectile is ___ m.

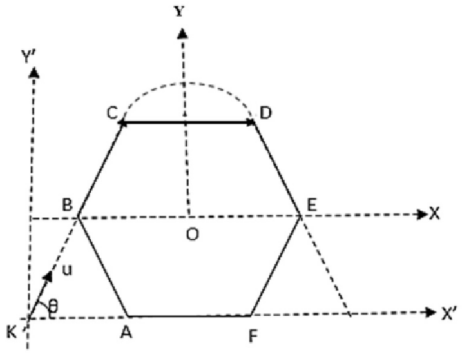
Wrong: Thonta Harshavardhan, K Ranveer Reddy, Mano Nischal Reddy D, Vihaan Bussari, Gajavali Anjaneyulu

Skipped: Dharshan Raj S, Poli Karthikeya Reddy, Shiva Sai Jyothiraditya Kokkonda

Q32 **Kinematics – Hexagon projectile – speed at the top**
Physics · Paragraph · marking +3 / 0 · marks lost by the top 10: 5 wrong × 3 + 3 skipped × 3 = 24

Lost by 8/10
5 wrong 3 skipped

PARAGRAPH-I



A regular hexagon of side a=$10\sqrt{3}$ m is shown in figure. From any point a particle is projected such that it touches B,C,D and E. The velocity of particle is v m/s. O is the centre of the hexagon. x-axis is parallel to horizontal and y-axis is perpendicular to AE and CD (g=10 m/s^2) $\left(\sqrt{3}=1.73\right)$.

32. The velocity of projectile at maximum height is _____ m/s

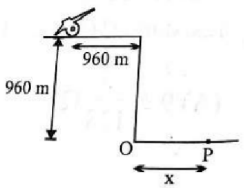
Wrong: Thonta Harshavardhan, K Ranveer Reddy, Mano Nischal Reddy D, Vihaan Bussari, Gajavali Anjaneyulu

Skipped: Dharshan Raj S, Poli Karthikeya Reddy, Shiva Sai Jyothiraditya Kokkonda

Q20 **Kinematics – Shell fired from a plateau – minimum landing distance**
Physics · Single correct · marking +3 / −1 · marks lost by the top 10: 4 wrong × 4 + 3 skipped × 3 = 25

Lost by 7/10
4 wrong 3 skipped

20. A gun is mounted on a plateau 960m away from its edge as shown. Height of plateau is 960m. The gun can fire shells with a velocity 100 m/s at any angle of the following choices. What is the minimum distance OP(=x) from the edge of plateau where shell of



gun can reach? (given g=10m/s^2, $\sin 74^0 = \sin 106^0 \simeq 0.96$)

A) 480 m B) 720 m C) 360 m D) 960 m

Wrong: Dharshan Raj S, Thonta Harshavardhan, Poli Karthikeya Reddy, Yapala Dhanush Reddy

Skipped: K Ranveer Reddy, Mano Nischal Reddy D, Gajavali Anjaneyulu

Q34 **Electrostatics – Flux through a cube from a wire along one edge**
Physics · Paragraph · marking +3 / 0 · marks lost by the top 10: 5 wrong × 3 + 1 skipped × 3 = 18

Lost by 6/10
5 wrong · 1 skipped

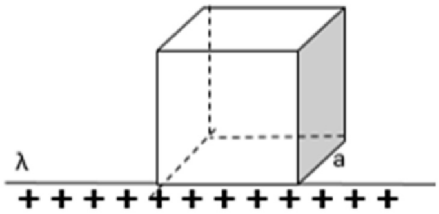
PARAGRAPH-II

The electric flux linked with a surface is the number of electric field lines associated with that area. Gauss law helps determine the flux

$$\oint \vec{E}.\vec{ds} = \frac{Q_{inc}}{\varepsilon_0}$$

For an open surface the flux will change depending on the solid angle. Considering this answer the following questions.

34.



A cube of edge length a contains an infinite wire of linear charge density λ at one of edges the flux associated with cube is given by $P\lambda a/\varepsilon_0$.The value of P upto two decimal places is :

Wrong: Thonta Harshavardhan, K Ranveer Reddy, Mano Nischal Reddy D, Gajavali Anjaneyulu, Yapala Dhanush Reddy **Skipped:** Poli Karthikeya Reddy

Q28 **Electrostatics – Flux through a cube with charges at its vertices**
Physics · Integer · marking +4 / 0 · marks lost by the top 10: 5 wrong × 4 = 20

Lost by 5/10
5 wrong · 0 skipped

28. A cube has charges each of value +Q placed at the vertices. The flux associated with the cube is $(X-1)\frac{Q}{\varepsilon_0}$, value of x is:

Wrong: Poli Karthikeya Reddy, K Ranveer Reddy, Shiva Sai Jyothiraditya Kokkonda, Vihaan Bussari, Gajavali Anjaneyulu **Skipped:** –

Q33 **Electrostatics – Flux through a ring – solid angle**
Physics · Paragraph · marking +3 / 0 · marks lost by the top 10: 5 wrong × 3 = 15

Lost by 5/10
5 wrong · 0 skipped

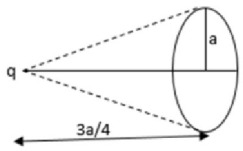
PARAGRAPH-II

The electric flux linked with a surface is the number of electric field lines associated with that area. Gauss law helps determine the flux

$$\oint \vec{E}.\vec{ds} = \frac{Q_{inc}}{\varepsilon_0}$$

For an open surface the flux will change depending on the solid angle. Considering this answer the following questions.

33.



The flux associated with the ring due to the charge q is given by Xq/ε_0. The value of X is approx.

Wrong: Poli Karthikeya Reddy, Shiva Sai Jyothiraditya Kokkonda, Vihaan Bussari, Gajavali Anjaneyulu, Yapala Dhanush Reddy

Skipped: –

Chemistry – high-impact questions

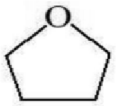
2 questions, worst first
cropped from the question paper

Q46 Alcohols, Phenols & Ethers – Identifying polar aprotic solvents

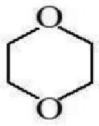
Chemistry · Integer · marking +4 / 0 · marks lost by the top 10: 5 wrong × 4 + 3 skipped × 4 = 32

Lost by 8/10
5 wrong 3 skipped

46. How many of the following are polar aprotic solvents?



Diethyl ether, Benzene, Ethyl alcohol,



Petrol

[P] [Q] [R] [S] [T] [U]

Acetone DMSO (dimethyl sulphoxide DMF (dimethyl formamide formic acid

[Q] [W] [X] [Y]

Wrong: Dharshan Raj S, Thonta Harshavardhan, K Ranveer Reddy, Mano Nischal Reddy D, Vihaan Bussari

Skipped: Shiva Sai Jyothiraditya Kokkonda, Gajavali Anjaneyulu, Yapala Dhanush Reddy

Q51 Carboxylic Acids – Moles of $LiAlH_4$ to reduce RCOOH to RCH_2OH

Chemistry · Paragraph · marking +3 / 0 · marks lost by the top 10: 5 wrong × 3 = 15

Lost by 5/10
5 wrong 0 skipped

PARAGRAPH-II

Carboxylic acids are organic compounds containing the carbonyl(>C=O) and hydroxyl (-OH) functional groups combined as a carboxyl group(-COOH). Due to resonance stabilization of the carboxylate anion, carboxylic acids exhibit higher acidity compared to alcohols and phenols. They participate in several characteristic reactions, including nucleophilic acyl substitution (esterification, amide formation, acid halide synthesis), decarboxylation, and reduction to primary alcohols. Furthermore, the presence of the –COOH group strongly influences physical properties, enabling extensive intermolecular hydrogen bonding that results in higher boiling points and stable dimeric structures in non-polar solvents.

51. The number of moles of $LiAlH_4$ required theoretically to reduce eight moles of carboxylic acid to primary alcohol

$$RCOOH \xrightarrow[2.H_3O^+]{1.LiAlH_4} RCH_2OH$$

Wrong: Dharshan Raj S, Thonta Harshavardhan, Poli Karthikeya Reddy, Shiva Sai Jyothiraditya Kokkonda, Gajavali Anjaneyulu

Skipped: –